

FRANK PORTER
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M1 MACRO

Group A

Session 2

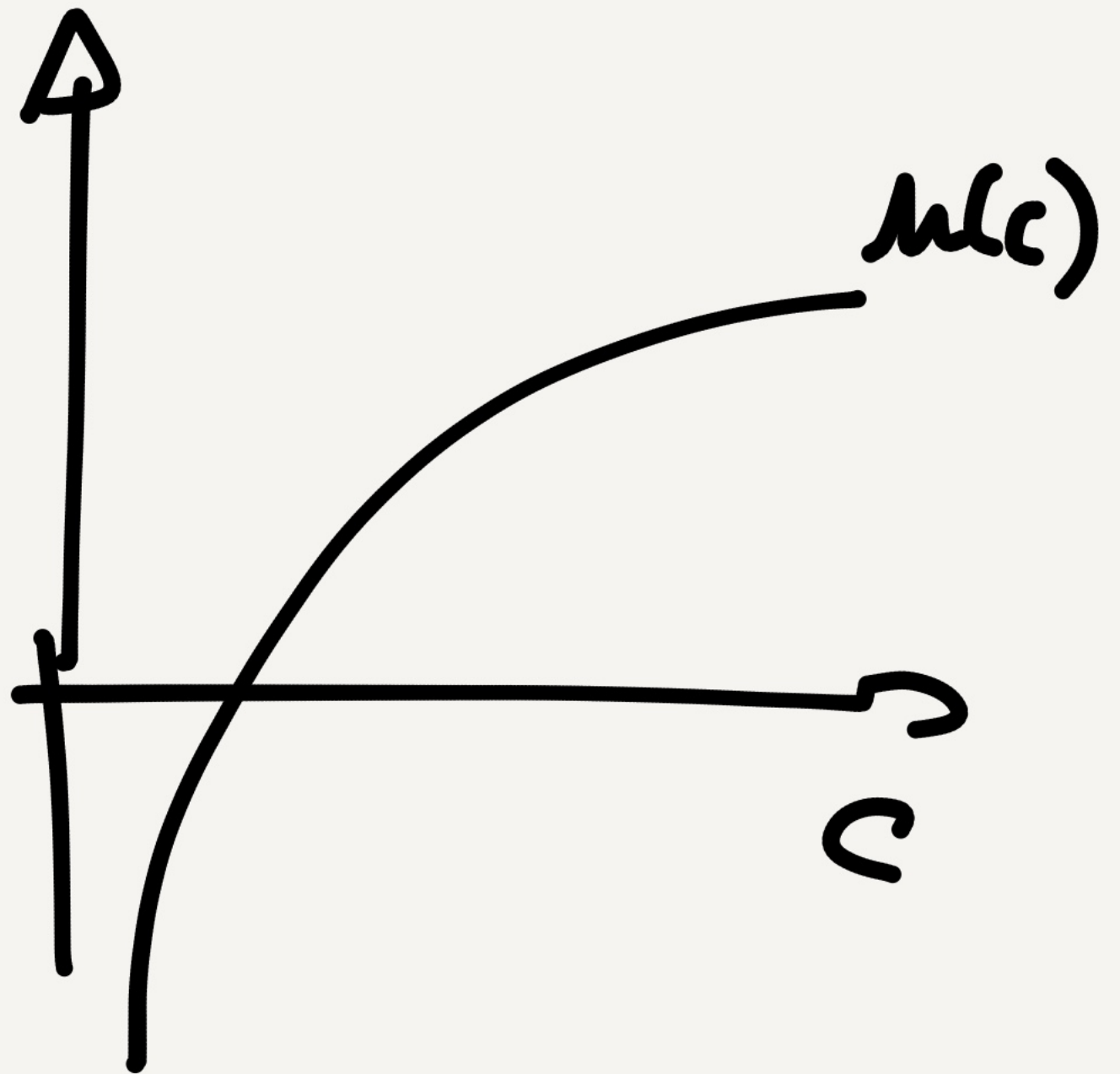
These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$\mu = \log c$$

$$\mu' = \frac{1}{c}$$

$$\mu'' = -\frac{1}{c^2} < 0$$

$$\mu' > 0$$



I solve for the gen. equilibrium of an exchange economy, meaning

① find optimal c_1, l_2 given P_1, P_2

② find P_1, P_2 such that all markets clear.

$$\textcircled{1} \quad \max \log C_1 + \beta \log C_2$$

$$\text{s.t.} \quad p_1 C_1 + p_2 C_2 = p_1 Y_1 + p_2 Y_2$$



$$C_2 = \frac{p_1 Y_1 + p_2 Y_2 - p_1 C_1}{p_2}$$

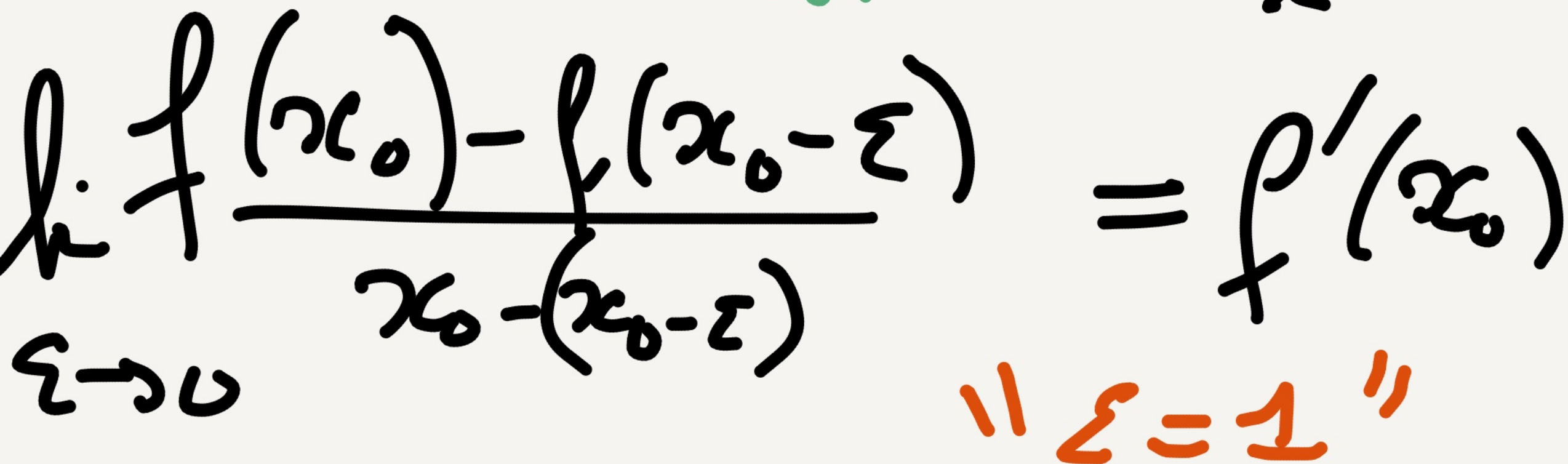
$$\Rightarrow \mu = \log C_1 + \beta \log \left[\frac{p_1 Y_1 + p_2 Y_2 - p_1 C_1}{p_2} \right]$$

$$\max \mu \text{ w.r.t } C_1 : \frac{d\mu}{dC_1} = 0 \quad \text{FOC}$$

Remark

$$f(x_0)$$

$$f(x_0 - 1) - f(x_0) \approx f'(x_0)$$



“ $\varepsilon = 1$ ”

Take a choice (c_1, c_2)

• decrease c_1 by 1 $\rightarrow$ utility is decreased by $u'(c_1) = \frac{1}{c_1}$

• to spend all the budget, I must

increase c_2 by $\frac{p_1}{p_2}$

$\rightarrow$ utility increase by $\beta \frac{1}{c_2} \times \frac{p_1}{p_2}$

$$C_1 - 1$$

$$C_2 + \frac{P_1}{P_2}$$

$$\text{loss } \frac{1}{C_1}$$

$$\text{gain: } \beta \frac{P_1}{P_2} \frac{1}{C_2}$$

if

$$<$$

$$=$$

Taking the full derivative of u
w.r.t. C_1 :

$$\frac{du}{dC_1} = \frac{1}{C_1} - \beta \frac{P_1}{P_2} \frac{1}{C_2} = 0$$

$\Rightarrow$

$$\frac{1}{C_1} = \beta \frac{P_1}{P_2} \frac{1}{C_2}$$

$$\Leftrightarrow \frac{\beta \frac{1}{c_2}}{\frac{1}{c_1}} = \frac{P_2}{P_1}$$

$$c_2 = \frac{\beta P_1 c_1}{P_2}$$

+ BC: $P_1 c_1 + P_2 c_2 = P_1 \gamma_1 + P_2 \gamma_2$

$$P_1 C_1 + P_2 \left(\beta \frac{P_1 C_1}{P_2} \right) = P_1 Y_1 + P_2 Y_2$$

$$P_1 (1 + \beta) C_1 = P_1 Y_1 + P_2 Y_2$$

$$C_1 = \frac{1}{1 + \beta} \left(Y_1 + \frac{P_2}{P_1} Y_2 \right)$$

$$C_2? \quad C_2 = \beta \frac{P_1}{P_2} C_1 \quad \Rightarrow$$

$$C_2 = \frac{\beta}{1+\beta} \left(\frac{P_1 Y_1}{P_2} + Y_2 \right)$$

② Therefore

Demand for good 1: C_1

(C_2)

Supply of —

Y_1

(Y_2)

Find (P_1, P_2) such that

$$C_1 = y_1$$

$$C_2 = y_2$$

$$C_1 = y_1$$

$$\Leftrightarrow \frac{1}{1+\beta} \left(y_1 + \frac{P_2}{P_1} y_2 \right) = y_1$$

$$y_1 + \frac{P_2}{P_1} y_2 = (1+\beta) y_1 \Leftrightarrow \boxed{\frac{P_2}{P_1} = \beta \frac{y_1}{y_2}}$$

$$C_2 = \gamma_2$$

$$\Leftrightarrow \frac{\beta}{1+\beta} \left(\frac{P_1 \gamma_1}{P_2} + \gamma_2 \right) = \gamma_2$$

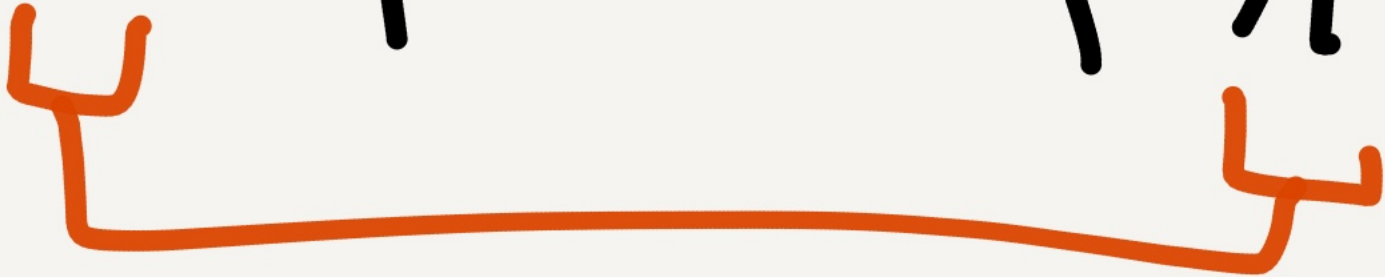
$$\frac{P_1}{P_2} \gamma_1 + \gamma_2 = \frac{1+\beta}{\beta} \gamma_2$$

$$\frac{P_1}{P_2} = \frac{1}{\beta} \frac{\gamma_2}{\gamma_1} \quad (\Rightarrow)$$

$$\frac{P_2}{P_1} = \beta \frac{\gamma_1}{\gamma_2}$$

$$C_1 = \gamma_1$$

Be $P_1 C_1 + P_2 C_2 = P_1 \gamma_1 + P_2 \gamma_2$



$$\Rightarrow P_2 C_1 = P_2 \gamma_2$$

$$\Leftrightarrow C_2 = \gamma_2$$

Walras law

If all agents are doing their budget constraints and if $n-1$ markets clear, then the n^{th} market will clear

$\Rightarrow$ I have only $n-1$ indep. equations

$\Rightarrow$ I can only $n-1$ prices

numéraire : the good whose
price is arbitrarily set to 1.

good 1 : $P_1 = \underline{1}$